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PnC Automation Tool - macOS Setup Guide

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REQUIREMENTS

- macOS 12 (Monterey) or newer
- Apple Silicon (M1/M2/M3/M4) or Intel Mac - both work
- Google Chrome installed (download from <https://www.google.com/chrome/>)
- Internet connection
- Your Mac admin password
- About 20-30 minutes for first setup

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PART A: ONE-TIME SETUP

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You only do this part once. Subsequent runs skip to Part B.

STEP 1: Extract the project

Extract the project zip somewhere easy to find. For example:

~/Desktop/Pnc_automation

(That's "Pnc_automation" as a folder inside your Desktop folder.)

Avoid putting it in:

- iCloud Drive folders (Documents/Desktop with iCloud sync on)
- Paths with weird characters or emoji
- Paths with spaces if you can help it

A simple ~/Desktop or ~/Projects works best.

STEP 2: Open Terminal

1. Press Cmd + Space to open Spotlight
2. Type: Terminal
3. Press Enter

A window opens with a prompt that looks something like:

```
yourname@MacBook ~ %
```

This is where you'll type commands. Everything below assumes you have this Terminal window open.

STEP 3: Navigate to the project folder

In Terminal, type this and press Enter (adjust the path if you extracted somewhere other than Desktop):

```
cd ~/Desktop/Pnc_automation
```

If you put the project somewhere else, easiest way to get the right path:

1. Open Finder
2. Find the project folder
3. Drag the folder into the Terminal window
4. Terminal types the full path for you
5. Add "cd " before it, press Enter

To confirm you're in the right place:

ls

You should see folders named:

- launcher
- university-job-portal
- vosyn-automation

Plus some .bat files (those are for Windows, ignore them).

If you see those, you're in the right place. Continue.

If you DON'T see those folders, you cd'd to the wrong location.

Try Finder-drag again.

STEP 4: Clear macOS download quarantine (one-time)

macOS marks files downloaded from the internet as "untrusted."

This can prevent the scripts from running. Clear that flag:

```
xattr -cr launcher/mac
```

This is a one-time fix per zip extraction. It's silent on success.

(If you get an error, it's safe to ignore - means there was nothing to clear.)

STEP 5: Make the scripts executable

Mac needs explicit permission to run shell scripts. Grant it:

```
chmod +x launcher/mac/setup_mac.sh
```

```
chmod +x launcher/mac/start_pnc_tool_mac.sh
```

```
chmod +x launcher/mac/stop_pnc_tool_mac.sh
```

These commands print no output. That's fine.

STEP 6: Run the setup script

```
./launcher/mac/setup_mac.sh
```

This takes 10-25 minutes depending on internet speed and what's already installed. It will:

1. Check for Google Chrome (fails fast if missing)
2. Install Homebrew if you don't have it
3. Install Python 3.12
4. Install Node.js 20
5. Create a Python virtual environment
6. Install all backend dependencies (~40 Python packages)
7. Install frontend dependencies (~hundreds of Node packages)

IMPORTANT WARNINGS THAT ARE NORMAL, NOT ERRORS:

- When Homebrew asks for your Mac password:
 - * You'll see "Password:" with a blinking cursor
 - * Type your password
 - * CHARACTERS WILL NOT APPEAR. This is normal Mac behavior.
 - * Press Enter when done.
- Homebrew prints A LOT of output (pages of "Pouring..." and "Installing..." messages). It is NOT stuck. Just wait.
- npm install also prints lots of warnings about "deprecated" packages. Those are not errors. Wait for it to finish.

When you see this at the end:

```
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Setup complete.
Now run:
./launcher/mac/start_pnc_tool_mac.sh
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```

Setup is done. Continue to Part B.

STEP 7 (only if Homebrew was just installed): Restart Terminal

If setup just installed Homebrew for you, future Terminal sessions won't know about it until you restart Terminal.

1. Close ALL Terminal windows
2. Open a fresh Terminal (Cmd + Space, "Terminal", Enter)
3. cd back to the project:
`cd ~/Desktop/Pnc_automation`

You're ready for Part B.

(If Homebrew was already installed, you can skip this step.)

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PART B: RUN THE APP
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Every time you want to use the tool, do this.

STEP 8: Start the app

From the project root in Terminal:

```
./launcher/mac/start_pnc_tool_mac.sh
```

You'll see:

- "Starting backend..." then "Backend is up."
- "Starting frontend..." then "Frontend is up."
- "Opening app..."
- Your default browser opens to <http://localhost:5173>

DO NOT CLOSE THE TERMINAL WINDOW while using the tool. The backend and frontend run inside this window. Closing it stops them.

(Unlike Windows, Mac runs everything in this one Terminal window.

Logs for backend and frontend stay in `/tmp/pnc-backend.log` and `/tmp/pnc-frontend.log` if you need to see them.)

STEP 9: Stop the app

In the Terminal window where you ran start, press:

Ctrl + C

The script will clean up both servers gracefully.

OR, from a different Terminal window: